Hitachi Deskstar HDS722525DLA380

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The Hitachi drives came with a bootable floppy containing the 'Hitachi Feature Tool', using which we could change the drive settings—from 1.5 Gbps to 3.0 Gbps—and even monitor the drive temperature.

Most switched-mode power supplies (SMPSes) have provision for the Molex power connector, which is the five-pin power connector used for CD-ROMs and IDE hard disks. Provision for a separate SATA power connector is usually absent in SMPSes, so including Molex power support along with a SATA power connector on the drive is a good thing. It would be ideal if manufacturers such as Seagate, Samsung and Maxtor had provided this.

WD's RAID Edition (Caviar RE2) had a few features unique to the maker, namely RAID-TLER and RAFF. (See box *Jargon Buster*.)

With the increase in capacity, the data density on the platter increases, and the accuracy of the head becomes important, especially in drives such as WD's Raptor (it's a 10,000 rpm disk). RAFF makes sure rotational vibrations are countered, so that the head reads data with minimal errors.

Based on support for both types of power connectors, additional features, and shock resistance, WD edged out the Hitachi drives to dictate terms in the Features section of the comparison.

Performance

The WD Raptor delivered outstanding scores in the synthetic tests, followed by other members of the WD family, and the lone Seagate. The Hitachi drives recorded the highest burst speeds in the HD-Tach benchmark. However, there's not much one can deduce from this, since the operating speed of the drive is much less than the burst speed. At the same time, this could have boosted Hitachi's performance in the real-world tests.

Synthetic Tests

HD-Tach

HD-Tach reports the burst speed of a drive in addition to other readings such as random and sequential read/write, which is common with the other benchmark tool—SiSoft Sandra. Maxtor's DiamondMax 10 trailed WD's Raptor by a small margin in average read and write speeds. Note that the maximum and minimum read/write speeds is just an outcome of the graph that HD-Tach plots while benchmarking; the highest peak is the maximum speed and the lowest dip is the minimum speed. Taking the average of these

How We Tested

To ensure that all the disks got equal treatment, we maintained a common test bed that would meet the requirements of all the disks. We installed Windows XP on our hard disk—a Seagate Barracuda 120 GB SATA. The test hard disk served, in each case, as a secondary drive. The test bed was powered by an AMD FX53 processor on an Asus A8N SLI Deluxe motherboard, with 1 GB of Corsair DDR RAM and an ATi Radeon X850 XT 256 MB Platinum Edition.

Features

Among the nine listed features, points were only given to the type of power connector, additional features, and shock resistance, as the rest of the features contributed to the Performance and Cost Per GB factors. Most of the additional features—such as NCQ and Hot-Plug—were common to all the drives, and some (such as RAFF, TLER and TCQ) were specific to the manufacturer; only the Parallel ATA drives differed in additional features, so the considerable difference in scores reflected only in the case of the Samsung's parallel ATA drives. The variation in this segment was negligible among the other drives.

Disks that supported both types of power connectors—Molex and SATA—were awarded more points in the Features category.

Performance

Our performance tests were classified into the synthetic tests and the real-world tests. The synthetic tests included the HD-Tach and SiSoft Sandra benchmarking tools.

HD-Tach runs on a raw partition. We did not award points to the minimum and maximum read/write speeds, since the data from those readings was considered inconsistent. Points were therefore allotted to average read and write speeds, CPU usage, burst speeds and random access time.

We used SiSoft Sandra's File System benchmark to estimate the Drive Index, Sequential Read, Sequential Write, Random Read, Random Write and Access Time. The score for read and write operations are in MB per second, while Access Time is in milli seconds.

The real-world test included the file transfer and Photoshop image load tests. We created a 700 MB assorted folder containing files of various sizes and types, whereas a 700 MB video clip served as the sequential file. The inter-drive transfer results were obtained by recording the time taken for transferring 700 MB of assorted and sequential data between the test hard disk and our primary hard disk (the one that had the OS on it).

A RAM drive is a virtual hard disk partition created using software. A pre-defined (712 MB) amount of RAM was allocated for this purpose. The intra-drive test was weighted over the other transfer tests as it tests the effectiveness of the drive's buffer.

For the Photoshop image load test, we created image files of varying sizes by increasing the resolution of a multi-layered image (in the PNG format). We created two FAT32 partitions on each test hard disk. The path to Photoshop's scratch disk, which acts as a virtual RAM space on a hard disk's partition, was changed to the test sample's first partition, and the images were stored in the second partition of the same test sample. The time taken to open the images was recorded as the observation.

Price Index And Warranty

We compared the drives based on cost per GB, obtained by the ratio of price to formatted capacity. Drives that had a higher capacity at a lower price scored higher on the Price Index. Warranties were either three or five years, and points were awarded accordingly.

Picking The Winners

The scores from Features, Performance, Price Index and Warranty were scaled out of 100 with relevant weightages given to each. The product that scored the highest was awarded the *Digit* Best Buy Gold, and the one that trailed the Gold winner was adjudged the *Digit* Best Buy Silver.